

Rising Waters - Entity Relationship Diagram Documentation

Page 1 – Entity Relationship Documentation

The Rising Waters project is designed to monitor flood conditions and support emergency response. The main entity is User, who registers and reports flood information. Each report contains the water level, flood status, location, and reporting time. Every user can submit multiple flood reports. The Flood_Report entity stores all flood-related details. Each report is uniquely identified by Report_ID. The relationship between User and Flood_Report is one-to-many. This structure helps in tracking reports efficiently. The database ensures that every report belongs to a valid registered user.

Page 2 – Relationships and Data Flow

When a flood report is submitted, the system generates an Alert if the water level exceeds the predefined threshold. The Alert entity stores the alert type and alert generation time. Rescue teams receive these alerts and respond to affected locations. The Rescue_Team entity contains team information, contact number, and assigned area. Each rescue team can handle multiple emergencies. The Emergency entity records the response status for each flood report. Foreign keys connect reports, alerts, and rescue teams, maintaining data integrity and improving emergency coordination.

Page 3 – Benefits of the ER Diagram

The ER diagram provides a clear representation of the database structure for the Rising Waters project. It reduces data redundancy by organizing information into related entities. Primary and foreign keys maintain accurate relationships between tables. The design supports quick retrieval of flood reports and emergency information. It also improves system scalability for future enhancements. Rescue operations become more efficient through proper data management. Alerts are generated accurately based on flood reports. Overall, the ER model ensures reliable storage, faster processing, and effective flood management.